Lab Report: Model comparison on benchmarks

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 70 percent training data and the rest for testing. We trained each model for 100 epochs with a batch size of 32.

2 Results

Figure 1 shows the test accuracy by dataset.

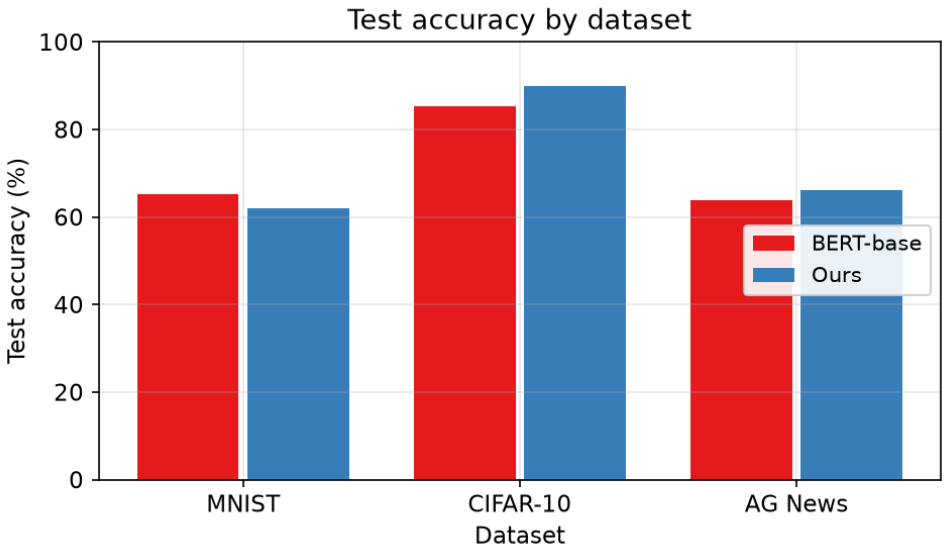


Figure 1: Test accuracy by dataset for the models in this experiment.

On AG News, Ours is 2.2 points better than BERT-base (Figure 1). On MNIST, Ours obtains a test accuracy of 71.9%.

Figure 2 shows the validation accuracy by dataset.

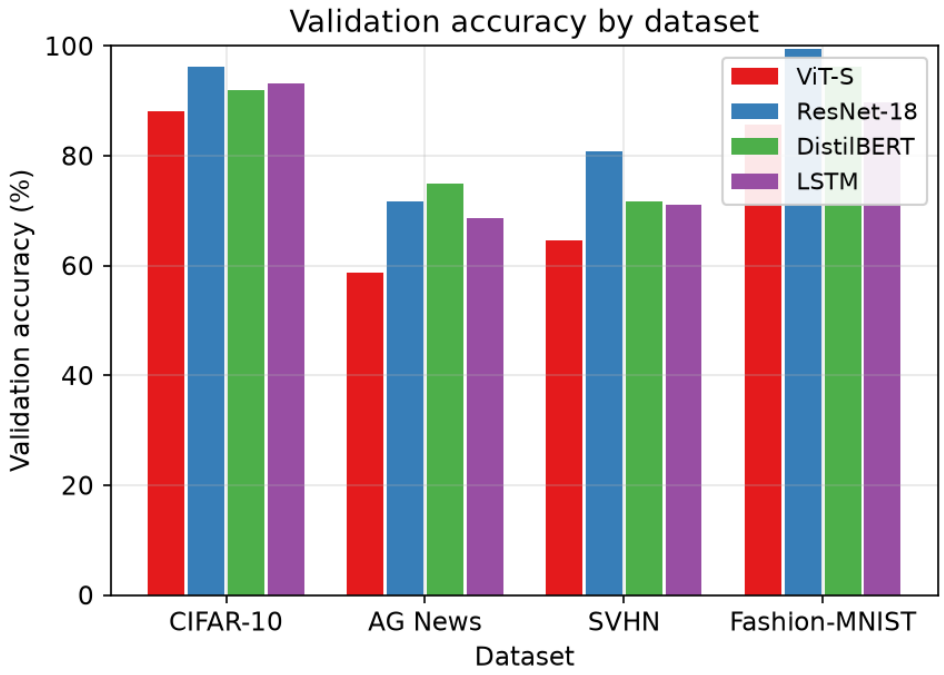


Figure 2: Validation accuracy by dataset for the models in this experiment.

ViT-S achieves 48.7% validation accuracy on AG News. As shown in Figure 2, LSTM outperforms ViT-S by 30.0 percentage points on AG News.

3 Discussion

XGBoost reaches 64.5% accuracy on Tiny-ImageNet. We note that the training was stable for all models.